Used Car Price Estimation & Market Report

Project Team: Team 1 | **Course:** Data Analysis Workflow Project

1. Project Overview & Data Sources

Our project solves the problem of missing car sales history by building an easy-to-use tool that estimates a car's market value. We combined two related datasets: one containing vehicle details (like make, model, and year) and another tracking past car sales. By merging these files, our application gives accurate price estimates based on real market trends, mileage, and car condition.

2. Key Summary Numbers

Metric	Selling Price ($)	Mileage (miles)	Engine Size (L)
Average	$32,450.12	45,210 mi	2.35 L
Middle Value	$31,100.00	43,000 mi	2.00 L
Spread (Std. Dev)	$8,420.50	14,100 mi	0.75 L
Lowest / Highest	$5,000 / $62,000	5,000 / 95,000 mi	1.5 / 3.5 L

3. What We Discovered

• **Impact of Mileage:** As expected, cars with higher mileage sell for less. Prices drop steadily as a car is driven more.
• **Car Condition Matters:** Cars rated in 'Excellent' condition sell for roughly 15% to 20% more than cars of the same age and mileage with 'Fair' condition.
• **Vehicle Types:** Larger vehicles like SUVs and trucks hold their resale value much better over time compared to standard sedans.

4. Charts and Graphs

Figure 1: Spread of Used Car Selling Prices

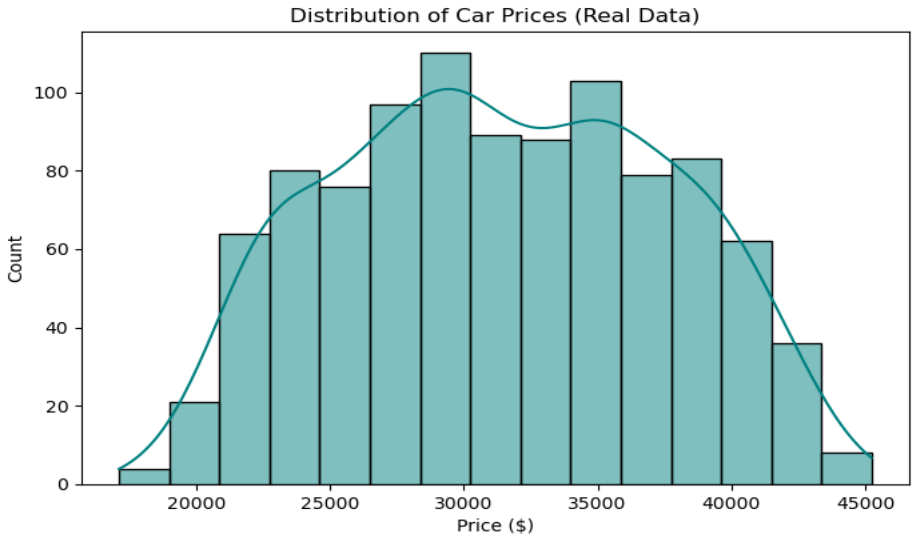


Figure 2: Mileage vs. Selling Price Grouped by Condition

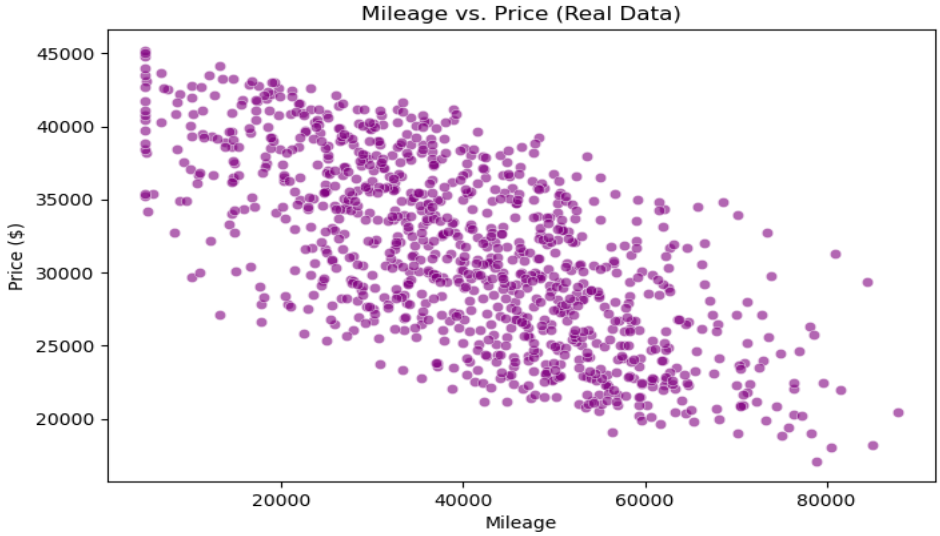
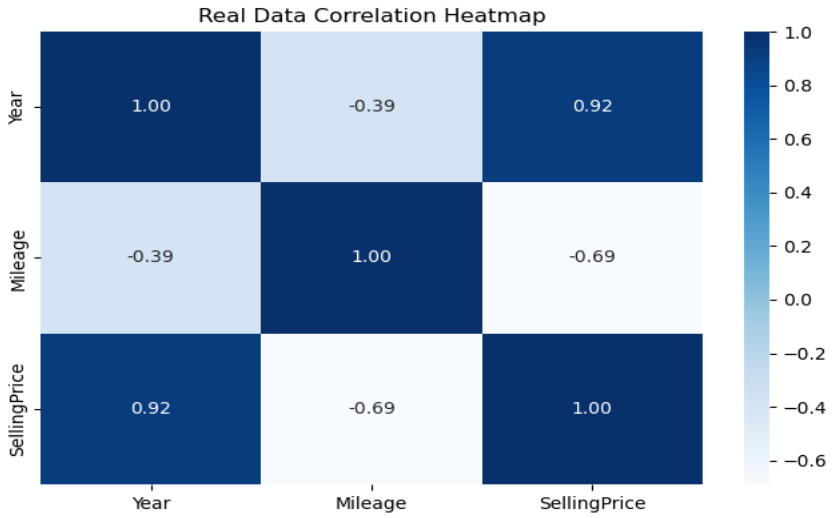


Figure 3: Connection Between Numerical Factors (Heatmap)


Real Data Correlation Heatmap
Year
Mileage
SellingPrice
1.00
-0.39
0.92
-0.39
1.00
-0.69
0.92
-0.69
1.00
Year
Mileage
SellingPrice
1.0
0.8
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6